

Ashenfall: Master World Map & Geological Specification

Complete Technical Data: Heightmaps, Coordinates, Lithosphere Splines, Climate Gradients & Still Life Population

Target Engine: Minecraft 1.21.1 (Pack 48) · Canvas: 8,000 × 8,000 Blocks · World: Continent of Vantyra · Author: Ashenfall Engineering

MASTER WORLDGEN BLUEPRINT: This document provides the definitive, production-grade specification for generating the finite 8,000 × 8,000 block continent of Vantyra. It establishes: (1) Exact 16-bit uint16 heightmap conversion math and Hermite continental shelf dropoff equations; (2) Spatial bounding boxes, target elevations, and coordinates for all 9 cardinal landmarks; (3) Lithosphere cubic spline density function mappings (continents, erosion, ridges); (4) The 5-dimensional multi-noise climate tensor, thermal lapse formulas, and anti-snow-pocket buffer rules; and (5) The 3 deterministic methodologies to force Still Life flora and feature decoration (fallen logs, mossy boulders, multi-tier canopies) across habitable sectors while preserving barren calderas and sheer cliffs.

1. Heightmap Data & Elevation Mathematical Specification

The continent of Vantyra operates on Minecraft 1.21.1's extended world height ceiling: **Y = -64 to Y = 320** (Total: 384 vertical blocks). Sea level is anchored at **Y = 62**.

16-Bit uint16 Normalization Equation: Every pixel in the master heightfield maps directly to a discrete block height through linear transformation:

```
h_norm = (Y - min_y) / (max_y - min_y) = (Y + 64.0) / 384.0
uint16_value = round(h_norm * 65535.0)
Y_block = -64.0 + (uint16_value / 65535.0) * 384.0
```

Geological Feature	Target Y Block	Norm [0..1]	16-Bit uint16	Lithosphere Layer / Material
Bedrock Floor	Y = -64.0	0.0000	0	Solid Bedrock Floor
Abyssal Trench (Veil of Salt)	Y = -32.0 to 10.0	0.0833 - 0.1927	5,461 - 12,629	Deep Cold Ocean / Gravel
Sunken Caldera Crater Floor	Y = 38.0 - 42.0	0.2656 - 0.2760	17,406 - 18,088	Basalt / Lava Basins / Netherite
Sunken Reach Drowned Shelf	Y = 52.0 - 56.0	0.3021 - 0.3125	19,798 - 20,480	Warm Ocean Reefs / Sandbars
Mean Sea Level (Water Line)	Y = 62.0	0.3281	21,504	Water Surface / Sea Surface
Forgotten Coast Shoreline	Y = 68.0 - 74.0	0.3438 - 0.3594	22,532 - 23,556	Stony Shore / Cold Pebble Beaches
Gilded Dunes Low Mesas	Y = 82.0 - 96.0	0.3802 - 0.4167	24,917 - 27,307	Terracotta / Red Sandstone
Cogwork March Quarry Benches	Y = 85.0 - 110.0	0.3880 - 0.4531	25,428 - 29,695	Terraced Badlands (9m steps)
Caldera Volcanic Rim	Y = 142.0 - 156.0	0.5365 - 0.5729	35,158 - 37,548	Blackstone / Basalt Spines
Subalpine Taiga Foothills	Y = 150.0 - 190.0	0.5573 - 0.6615	36,523 - 43,351	Podzol / Coarse Dirt (Pine Tree Limit)

Glacial Spine Arêtes & Horns	Y = 220.0 - 279.2	0.7396 - 0.8938	48,471 - 58,575	Frozen Peaks / Blue Glacier Ice
World Build Limit Ceiling	Y = 320.0	1.0000	65,535	Atmospheric Sky Ceiling

Continental Shelf Dropoff (Hermite S-Curve Formula): Beyond the continental shelf radius of $R_{\text{inner}} = 3,300$ blocks, terrain descends monotonically into the abyss, reaching maximum oceanic depth at $R_{\text{outer}} = 3,800$ blocks. The continuous dropoff is formulated as:

```
t = clamp((radius - 3300.0) / (3800.0 - 3300.0), 0.0, 1.0)
S(t) = 3.0 * t^2 - 2.0 * t^3 # Continuous Cubic Hermite Spline
H_drop(R) = -600.0 * S(t) # Deep ocean dropoff plunging into the Veil of Salt
```

2. Landformations & Exact Coordinate Specifications

The continent of Vantyra is centered at $(X=0, Z=0)$, spanning exactly $-4,000$ to $+4,000$ blocks along both horizontal axes. The 9 cardinal landmarks are fixed at the following designated coordinates:

Landmark Name	Center Coord	Bounding Box (X1..X2, Z1..Z2)	Target Y	Geological Formation Rules
1. Forgotten Coast (Spawn)	(0, 68, 2500)	X: [-600, 600] Z: [2000, 3200]	68 - 74	Cold pebble bluffs, rolling wildflower bluffs, ancient stone beacon, starter pier.
2. Cogwork March	(-2100, 85, 0)	X: [-2800, -1400] Z: [-700, 700]	85 - 110	Concentric 9m quarry steps, brass river chasms, Create factory benches.
3. The Ashen Caldera	(0, 80, 0)	X: [-750, 750] Z: [-750, 750]	38 - 150	Volcanic ring wall (Y=146), sunken crater basin (Y=40), Obsidian Throne spire (Y=92).
4. Solitary Glacial Spine	(0, 220, -2500)	X: [-1800, 1800] Z: [-3500, -1500]	180 - 279	Alpine cordillera, Matterhorn arêtes, cirque tarns, flash-frozen pilgrim trails.
5. The Gilded Dunes	(2300, 75, 0)	X: [1600, 3100] Z: [-800, 800]	75 - 94	Vitrified black-glass dunes, 45° barchan ridges, terracotta canyon mesas.
6. The Whispering Fen	(2000, 63, 2000)	X: [1300, 2700] Z: [1300, 2700]	62 - 66	Flat sunken bayous, braided delta channels, giant fungal heartwood trees.
7. The Sunken Reach	(-2400, 54, 1600)	X: [-3200, -1700] Z: [1000, 2300]	50 - 62	Drowned coastal caldera shelf (100 fathoms), coral atolls, barrier sandbars.
8. The Hermit's Spire	(-1800, 140, -1800)	X: [-2300, -1300] Z: [-2300, -1300]	140 - 185	Solitary granite needles, ascetic rope bridges, silent wind-scoured cloisters.
9. The Byzantine Choir	(1800, 120, -1800)	X: [1300, 2300] Z: [-2300, -1300]	110 - 145	Gilded resonant basilicas, pink cherry terraces, harmonic acoustic ruins.
RIM: The Veil of Salt	Radius > 3550	Full perimeter to [-4000, 4000]	-32 - 62	Syrupy caustic brine abyss, white salt crusts, space unravels beyond 4000.

3. Lithosphere Biomes & Spline Density Function Architecture

Lithosphere entirely replaces vanilla piecewise linear interpolation with **Continuous Cubic Hermite Splines** ($S(t) = 3t^2 - 2t^3$). The macro-terrain is governed by three primary density functions in datapacks/ashenfall_data2:

- continents.json:** Continuous cubic spline mapping continentalness $C \in [-1.2, 1.2]$. Ocean basins are clamped below $C < -0.20$; landmass emerges smoothly at $C \geq 0.05$.
- erosion.json:** Multi-octave domain-warped spline ($E \in [-1.0, 1.0]$). Low erosion ($E < -0.45$) produces jagged alpine arêtes; high erosion ($E > 0.35$) produces flat floodplains and bayous.
- ridges.json:** Folding harmonic spline ($R \in [-1.0, 1.0]$). Carves wide U-shaped river valleys at $R \approx 0$ while

cresting sharp ridges at \$|R| \approx 1\$.

Region	Target Lithosphere Biomes	Cont (C)	Erosion (E)	Ridges (R)	Surface Blocks (Lithosphere)
Forgotten Coast	minecraft:plains minecraft:meadow	0.10 - 0.30	-0.10 - 0.20	-0.30 - 0.30	Grass Block, Podzol, Stony Shore
Glacial Spine	minecraft:frozen_peaks minecraft:jagged_peaks minecraft:grove	0.45 - 0.90	-0.80 - -0.45	0.50 - 0.95	Snow Block, Packed Ice, Calcite, Stone
Cogwork March	minecraft:windswept_hills minecraft:wooded_badlands	0.25 - 0.60	-0.35 - 0.15	-0.80 - -0.20	Orange/Yellow Terracotta, Stone, Andesite
Ashen Caldera	minecraft:basalt_deltas minecraft:eroded_badlands	0.20 - 0.50	-0.20 - 0.30	0.00 - 0.40	Basalt, Blackstone, Magma Block, Obsidian
Gilded Dunes	minecraft:desert minecraft:badlands	0.20 - 0.55	0.25 - 0.70	-0.40 - 0.40	Red Sand, Sandstone, Terracotta
Whispering Fen	minecraft:swamp minecraft:mangrove_swamp	0.05 - 0.25	0.40 - 0.85	-0.20 - 0.20	Mud, Peat, Moss, Coarse Dirt
Sunken Reach	minecraft:warm_ocean minecraft:lukewarm_ocean	-0.35 - -0.10	0.10 - 0.50	-0.50 - 0.50	Sand, Coral Reefs, Prismarine Gravel
Veil of Salt (Abyss)	minecraft:deep_cold_ocean minecraft:deep_ocean	-0.90 - -0.45	-0.60 - 0.40	-1.00 - 1.00	Gravel, Deepslate, Packed Salt

4. Temperature Gradient, Altitude Lapse Rates & Biome Blending

The 5-Tier Climate Separation Rule: Previous worldgen builds suffered from the 'snow-pocket glitch'—isolated patches of frozen snow generating inside temperate forests. This occurred because temperate and freezing biomes were separated by only 0.10 temperature units, allowing noise jitter to flip the Voronoi cell. In Ashenfall, climate is governed by strict **0.55-unit temperature buffering**:

Climate Tier	Normalized Temp (T)	Humidity (H)	Target Geography & Buffer Behavior
Tier 0: Glacial Arctic	T <= -0.75	H: [-0.4, 0.4]	Glacial Spine summits (Y > 180). Frozen peaks, ice sheets, permafrost.
Tier 1: Boreal Buffer Belt	-0.60 <= T <= -0.25	H: [-0.2, 0.6]	Mandatory 600-block non-snowy pine taiga buffer. Prevents snow from touching temperate plains.
Tier 2: Temperate Lowlands	-0.15 <= T <= 0.40	H: [-0.1, 0.8]	Forgotten Coast, plains, meadows, oak/birch woodlands.
Tier 3: Subtropical Bayou	0.35 <= T <= 0.60	H: [0.5, 1.0]	Whispering Fen. Saturated peat, mangrove deltas, warm swamps.
Tier 4: Arid & Volcanic	T >= 0.70	H: [-0.8, -0.2]	Gilded Dunes and Ashen Caldera. Desert sand seas, terracotta badlands, basalt sinks.

Atmospheric Altitude Lapse Rate Formula: Temperature decreases linearly with elevation above sea level (\$Y = 62\$), simulating genuine adiabatic expansion:

```
# Effective Temperature = Latitude Base + Altitude Lapse Rate
T_eff(x, y, z) = T_base(x, z) - 0.0055 * max(0.0, y - 62.0)
# High summits at Y=220 experience a -0.869 temperature reduction, guaranteeing snow caps.
```

5D Voronoi Metric & Biome Blending: Biome placement evaluates the Euclidean distance in 5D multi-noise space, smoothed across an 8-block Voronoi cell radius:

```
D(P, B_i) = sqrt((T - T_i)^2 + (H - H_i)^2 + (C - C_i)^2 + (E - E_i)^2 + (W - W_i)^2)
biome_blend_radius = 4 # 9x9 chunk voxel smoothing in Minecraft client options
```

5. Population Instructions: How to Force Still Life to Generate Where Desired

Still Life features (branched canopy trees, fallen birch/oak logs, glacial erratic boulders, and wildflower carpets) do not rely on random brush strokes. They are governed by three deterministic mechanisms that can be programmatically forced:

Method 1: The WorldPainter 'Populate' Layer (100% Guaranteed Native Decorator Pass):
When pre-generating terrain via WorldPainter or Python, never export chunks as Status: 'full'. Instead, apply the binary ASHFALL_POPULATE_MASK.png to WorldPainter's native **Populate** layer. When Minecraft loads a chunk flagged with Populate, its internal world decorator natively executes Still Life's entire placed feature registry. Valleys receive lush trees and fallen logs; cliffs and volcanic calderas remain completely bare.

Method 2: Still Life Biome Tag Injections (Datapack Level):
Still Life feature placement is bound to biome tags in data/still_life/tags/worldgen/biome/. By adding target biomes into these tags, Still Life is forced to generate in those specific biomes:

```
# data/still_life/tags/worldgen/biome/has_canopy.json -> ["minecraft:plains", "minecraft:forest"]
# data/still_life/tags/worldgen/biome/has_fallen_logs.json -> ["minecraft:forest", "minecraft:taiga"]
# data/still_life/tags/worldgen/biome/has_boulders.json -> ["minecraft:windswept_hills", "minecraft:meadow"]
```

Method 3: The Still Life Slope-Aware Surface Rule:
Still Life inspects local geological slope angles. Tree and log decorators require grass/dirt surface blocks. Terrain steeper than 35° automatically strips topsoil to bare rock scree, naturally preventing trees from generating on cliff faces:

Slope Angle (θ)	Surface Block (Still Life Rule)	Vegetation & Feature Population Density
0° <= θ < 25°	Grass Block / Deep Loam (3-5 blocks)	100% Full Canopy Trees, Fallen Logs, Wildflowers, Shrub understory.
25° <= θ <= 35°	Coarse Dirt, Permadirt, Podzol	40% Density: Pine Taiga, berry bushes, small stone erratics.
35° < θ <= 45°	Cobblestone, Stone Scree, Gravel	5% Density: Mossy boulders, clinging lichen. Trees strictly excluded.
θ > 45° (Sheer Cliffs)	Solid Bedrock, Granite, Basalt	0% Population: 100% bare rock scree. No trees or logs can spawn.
Y > 225.0 (Treeline)	Snow Block, Packed Ice, Calcite	0% Population: Climatic treeline cutoff. High-altitude glacial frost.
Caldera Rim / Crater	Blackstone, Basalt, Magma Block	0% Population: Volcanic barrenness mask overrides all foliage.

Turnkey Execution Command: To re-generate all heightmaps, 16-bit uint16 assets, Still Life populate masks, and JSR-223 WorldPainter scripts, run:

```
python generate_ashfall.py --res 2048 --out worldpainter
# Output: ASHFALL_HEIGHTMAP_16BIT.png, ASHFALL_POPULATE_MASK.png, ashenfall_worldpainter_setup.js
```